Abdelrahman Hazem Khalil

Mechatronics Engineer

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Arabic: Native speaker | English: Intermediate

Profile

Mechatronics Engineer with hands-on experience in machine vision, robotics, automation, and AI integration. Developed and deployed industrial vision inspection systems using smart cameras, image processing, and C# interfaces, alongside ROS 2-based robotic systems and embedded automation projects. Experienced in combining hardware and software to build practical engineering solutions, with a strong interest in machine vision, AI, and intelligent automation.

Education

Bachelor of Engineering (Hons.) in Mechatronics Engineering

Asia Pacific University of Technology & Innovation, Kuala Lumpur, Malaysia | Graduated Oct 2025

Master of Mechatronics Engineering

De Montfort University, UK

(Collaborative programme with Asia Pacific University) | 2021-2025

Technical Skills

- **Programming Languages:** C/C++, Python, C#, MATLAB
- **Embedded Systems & Robotics:** ESP32, Arduino, Microcontrollers, ROS 2, Motor Control, Sensor Integration, Gazebo, RViz, SLAM
- **Computer Vision & AI:** OpenCV, MediaPipe, Zebra Smart Cameras (VS40/FS10), Vision-Language Models (VLMs), Custom Disease/Defect Detection Models
- **Tools & Software:** SolidWorks, Automation Studio, CX Programmer & Designer, AutoCAD

Experience

Project Engineer Intern

Control Easy Technology, Malaysia | Feb 2024 – Jun 2024

- Designed, calibrated, and deployed automated quality control vision systems on active production lines using Zebra VS40 and FS10 smart cameras.
- Engineered an automated ECU scratch and bent-pin detection scanner capable of identifying sub-0.2mm pin deviations; optimized lighting setups and thresholding logic to eliminate false positives caused by factory vibration and shadows.
- Programmed C# graphical user interfaces to interface directly with camera servers for real-time image processing, data collection, and equipment control.
- 3D-modeled, printed, and tested 5 iterations of custom universal camera mounting hardware in SolidWorks to ensure structural stability in high-vibration environments.

ICT Draftsman

AVIS, Egypt | Dec 2025 – Jul 2026

- Reviewed architectural layouts and designed low-current/ICT infrastructure and Project Design Matrices (PDMs) for large-scale developments, including hotels, hospitals, and residential projects.
- Mapped multi-story network connectivity, calculating placements for floor-level racks, patch panels, data outlets, and single-line diagrams.
- Engineered an internal AI-powered symbol-counting tool utilizing Vision-Language Models (VLMs) and automation scripts to accelerate floor plan drafting workflows.
- Developed a RAG-based application to search technical specification PDFs and retrieve specific project requirements, including spare capacity percentages and equipment allowances.

Projects

- ELA 2.0 AI-Powered Humanoid Robot (FYP): Developed a ROS 2-based humanoid robot with autonomous navigation, voice interaction, and gesture recognition. Integrated MediaPipe and RViz for robotic arm tracking across variable lighting environments.
- Smart AGV Chili Farm (Group Design Project): Led the machine vision subsystem to detect crop diseases using custom vision models; designed a targeted,

automated spraying mechanism and assisted in AGV drive system and electronics integration.

- Automatic Stamping System: Programmed and simulated an automated stamping sequence utilizing CX Programmer (PLC) and CX Designer (HMI) for production line integration.
- Automated Storage Elevator & Mechanical Systems: Designed and implemented a pneumatic cylinder-driven storage elevator for sequential motion control. Performed full mechanical analysis (torque, power, material selection) for a meat blender design.
- Eva AI Desktop Assistant: Developed a natural language desktop assistant using local language models (Mistral) for task automation, email processing, and system workflows

More Information
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